

# Predicting Superconducting Critical Temperature from Engineered Material Descriptors: A Comparative Machine Learning Study

Israfil Saygitov\*

Nursan Omarov†

## Abstract

Superconducting materials, which conduct electric current with zero resistance below a critical temperature ( $T_c$ ), represent one of the most promising avenues for reducing energy losses in electrical infrastructure. Identifying material compositions associated with a higher  $T_c$  remains a central open problem in condensed matter physics and materials science. This study investigates whether machine learning models can predict  $T_c$  from a set of 81 engineered numerical descriptors derived from elemental and compositional properties. Six regression models — Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and a deep multilayer perceptron (MLP) — were trained and evaluated on a public superconductivity dataset (21,263 samples after cleaning) using an 80/20 train–test split and 5-fold cross-validation. Random Forest achieved the strongest performance (cross-validated RMSE  $\approx 9.70$  K, test  $R^2 \approx 0.930$ ), narrowly ahead of XGBoost (CV-RMSE  $\approx 9.88$  K), while the deep MLP placed in the middle of the field (test  $R^2 \approx 0.879$ ) and Linear Regression performed weakest ( $R^2 \approx 0.738$ ), indicating a strongly nonlinear relationship between descriptors and  $T_c$  on which tree ensembles remain especially strong. Impurity-based and permutation feature importance agree that thermal-conductivity-, atomic-mass-, and valence-related descriptors are the most informative predictors, and Principal Component Analysis confirms that the dataset contains non-random structure related to  $T_c$ . We additionally quantify a data-leakage risk: 28.7% of feature rows are exact duplicates, and removing them lowers Random Forest performance to  $R^2 \approx 0.911$ , so the leakage-controlled figure is the more conservative estimate. We discuss the limitations of this proof-of-concept pipeline and outline directions for developing it into a screening tool for candidate superconducting materials.

*Keywords:* superconductivity; critical temperature; machine learning; random forest; XGBoost; deep learning; feature importance; materials informatics.

## 1. Introduction

### 1.1 Motivation

In 2026, severe flooding struck the Republic of Dagestan, Russia, and local officials attributed part of the resulting infrastructure damage to degraded electrical wiring — cables that wear out, overheat, and fail under stress. This observation motivated the present study: what if electrical grids did not lose energy to resistance at all?

Superconductors are materials that conduct electric current with zero electrical resistance, meaning they do not dissipate energy as heat and, in principle, do not degrade through resistive overheating in the way conventional conductors do. Widespread use of superconducting materials in

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\*Independent Researcher. Corresponding author.

†Independent Researcher.

power infrastructure could substantially improve grid reliability and efficiency. The central obstacle to this vision is that most known superconductors only exhibit this property at extremely low temperatures, which makes them impractical for large-scale real-world deployment.

Identifying materials with a higher critical temperature ( $T_c$ ) — the temperature below which superconductivity occurs — is therefore one of the most important open problems in modern condensed matter physics and materials science. This work applies machine learning methods to predict  $T_c$  from engineered material descriptors, as an initial step toward a computational screening pipeline for promising superconducting candidates.

## 1.2 Objectives

The specific objectives of this study are to: (i) compare the predictive performance of six regression algorithms — including a deep neural network — on a public superconductivity dataset; (ii) identify which engineered descriptors are most informative for predicting  $T_c$ ; and (iii) assess, through correlation analysis, dimensionality reduction, and a duplicate-leakage check, whether the dataset exhibits structure consistent with a learnable physical relationship and how robust the reported performance is.

# 2. Data and Methods

## 2.1 Dataset

The dataset used in this study consists of engineered numerical descriptors for superconducting materials, together with the corresponding critical temperature (`critical_temp`) as the target variable. Each descriptor is derived from elemental and compositional properties of the constituent atoms (e.g., atomic mass, valence, thermal conductivity, electron affinity, atomic radius, and fusion heat), aggregated using statistics such as mean, weighted mean, geometric mean, standard deviation, range, and entropy across the elements present in each material. The raw dataset contained 21,263 usable records after preprocessing, with 81 descriptor features. The data originate from the UCI Machine Learning Repository superconductivity dataset of Hamidieh [1].

## 2.2 Preprocessing

One column, `wtd_std_Density`, was coerced to a numeric type, and any rows with resulting invalid or missing values were removed. The target variable was set to `critical_temp`, and all remaining numerical columns were used as input features. The cleaned dataset was split into training (80%) and test (20%) subsets using a fixed random seed to ensure reproducibility.

## 2.3 Models

Six regression models were trained and compared:

- **Linear Regression**, as a baseline linear model.
- **Decision Tree Regressor**, a single nonlinear tree-based model.

- **Random Forest Regressor** (300 estimators), an ensemble of bagged decision trees.
- **Gradient Boosting Regressor**, a sequential boosting ensemble.
- **XGBoost Regressor** (300 estimators, max depth 6, learning rate 0.05, `subsample` 0.8, `colsample_bytree` 0.8), a regularized gradient boosting implementation.
- **Multilayer Perceptron (MLP)**, a deep feed-forward neural network implemented in PyTorch, with three hidden layers (256–128–64 units), batch normalization, ReLU activations, and dropout regularization. Inputs and target were standardized using training-set statistics, and the network was trained with the Adam optimizer under mean-squared-error loss with early stopping on an internal validation split.

All models were evaluated on the held-out test set using mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and the coefficient of determination ( $R^2$ ). To assess the stability of each model’s performance, 5-fold cross-validation was additionally performed on the training set, reporting the mean and standard deviation of the cross-validated RMSE (CV-RMSE, CV-STD).

## 2.4 Feature Importance and Structural Analysis

To interpret the best-performing model, impurity-based feature importances were extracted from the Random Forest model. Because impurity-based importances can be biased when features are correlated, permutation importance was additionally computed on the held-out test set as a robustness check. In parallel, Pearson correlation coefficients between each descriptor and `critical_temp` were computed across the full cleaned dataset. Finally, Principal Component Analysis (PCA) was applied to the standardized feature matrix to visualize the two-dimensional structure of the dataset with respect to critical temperature.

## 2.5 Duplicate and Leakage Analysis

Engineered superconductivity descriptors are known to contain many near-identical compositions. Because a random train–test split can place near-duplicate materials in both partitions and leak information, we counted exact-duplicate feature rows and re-evaluated the best model on a deduplicated version of the dataset as a conservative robustness check.

# 3. Results

## 3.1 Model Comparison

Table 1 summarizes the performance of all six models on the test set and under 5-fold cross-validation. Random Forest achieved the best overall performance (test MAE = 5.08 K, RMSE = 8.97 K,  $R^2$  = 0.930; CV-RMSE =  $9.70 \pm 0.25$  K), closely followed by XGBoost (CV-RMSE =  $9.88 \pm 0.28$  K); the two tree ensembles are statistically very close. The deep MLP placed in the middle of the field (test  $R^2$  = 0.879, CV-RMSE =  $12.17 \pm 0.23$  K), outperforming a single Decision Tree and Gradient Boosting on cross-validated RMSE but not matching the leading tree ensembles.

**Table 1:** Performance comparison of six regression models on the test set, sorted by cross-validated RMSE.

| Model               | MAE    | MSE     | RMSE   | $R^2$ | CV-RMSE | CV-STD |
|---------------------|--------|---------|--------|-------|---------|--------|
| Random Forest       | 5.076  | 80.452  | 8.970  | 0.930 | 9.700   | 0.253  |
| XGBoost             | 5.811  | 86.037  | 9.276  | 0.925 | 9.876   | 0.280  |
| MLP (Deep Learning) | 7.697  | 139.165 | 11.797 | 0.879 | 12.173  | 0.226  |
| Decision Tree       | 6.090  | 139.449 | 11.809 | 0.879 | 12.539  | 0.296  |
| Gradient Boosting   | 8.501  | 152.150 | 12.335 | 0.868 | 12.794  | 0.149  |
| Linear Regression   | 13.211 | 302.008 | 17.378 | 0.738 | 17.699  | 0.253  |

Linear Regression performed markedly worse than all nonlinear models ( $R^2 = 0.738$ , CV-RMSE = 17.70 K).

The gap between Linear Regression and the tree-based ensembles suggests that the relationship between the engineered descriptors and critical temperature is strongly nonlinear. Notably, the deep MLP — despite its capacity to model nonlinear interactions — did not surpass Random Forest or XGBoost, consistent with the broader observation that gradient-boosted and bagged tree ensembles remain strong baselines for tabular data with heterogeneous engineered features.

### 3.2 Feature Importance

Table 2 lists the descriptors with the highest impurity-based importance in the Random Forest model. A single feature, `range_ThermalConductivity`, dominates the importance ranking (0.536), followed by `wtd_gmean_ThermalConductivity` (0.126). The remaining top features relate to atomic mass, valence, and density, together accounting for a much smaller share of total importance.

**Table 2:** Top 8 features by Random Forest impurity-based importance.

| Feature                                      | Importance |
|----------------------------------------------|------------|
| <code>range_ThermalConductivity</code>       | 0.536      |
| <code>wtd_gmean_ThermalConductivity</code>   | 0.126      |
| <code>std_atomic_mass</code>                 | 0.022      |
| <code>wtd_gmean_Valence</code>               | 0.019      |
| <code>std_Density</code>                     | 0.011      |
| <code>mean_Density</code>                    | 0.011      |
| <code>wtd_entropy_ThermalConductivity</code> | 0.011      |
| <code>gmean_ElectronAffinity</code>          | 0.011      |

We note that this pronounced concentration of importance in thermal-conductivity-related descriptors may partly reflect multicollinearity among related features (e.g., `range_`, `wtd_gmean_`, and `wtd_std_ThermalConductivity` are derived from the same underlying property) rather than a single dominant physical mechanism; this is discussed further in Section 4.

### 3.3 Permutation Importance

To guard against the known bias of impurity-based importance under feature correlation, permutation importance was computed on the held-out test set (Table 3). The two measures

agree closely: `range_ThermalConductivity` remains dominant ( $0.722 \pm 0.019$ ), followed by `wtd_gmean_ThermalConductivity`, `std_atomic_mass`, and `wtd_gmean_Valence`. The agreement between two independent importance measures strengthens confidence that these descriptors carry genuine predictive signal rather than being artifacts of a single scoring method.

**Table 3:** Top 8 features by Random Forest permutation importance (mean  $\pm$  std over 5 repeats, held-out test set).

| Feature                                    | Permutation importance |
|--------------------------------------------|------------------------|
| <code>range_ThermalConductivity</code>     | $0.722 \pm 0.019$      |
| <code>wtd_gmean_ThermalConductivity</code> | $0.079 \pm 0.005$      |
| <code>std_atomic_mass</code>               | $0.039 \pm 0.002$      |
| <code>wtd_gmean_Valence</code>             | $0.033 \pm 0.001$      |
| <code>mean_Density</code>                  | $0.018 \pm 0.001$      |
| <code>entropy_FusionHeat</code>            | $0.015 \pm 0.001$      |
| <code>entropy_Density</code>               | $0.014 \pm 0.000$      |
| <code>range_atomic_radius</code>           | $0.014 \pm 0.001$      |

### 3.4 Correlation Analysis

Pearson correlation coefficients between individual descriptors and `critical_temp` ranged from approximately  $-0.63$  to  $+0.72$ . The strongest positive correlations were observed for `wtd_std_ThermalConductivity` ( $r = 0.721$ ), `range_ThermalConductivity` ( $r = 0.688$ ), `range_atomic_radius` ( $r = 0.654$ ), and `std_ThermalConductivity` ( $r = 0.654$ ). The strongest negative correlations were observed for `wtd_mean_Valence` ( $r = -0.632$ ) and `wtd_gmean_Valence` ( $r = -0.616$ ). These results are broadly consistent with the Random Forest feature importance ranking and indicate that thermal-conductivity-, radius-, and valence-related descriptors carry meaningful information about critical temperature, although correlation alone does not establish causal physical relationships.

### 3.5 Distribution and Structure of the Target Variable

The distribution of `critical_temp` across the dataset is right-skewed (skewness  $\approx 0.86$ ; mean  $\approx 34.4$  K, median  $\approx 20.0$  K, maximum 185 K): most materials exhibit relatively low critical temperatures, with a smaller number of materials reaching substantially higher values. This asymmetry means the RMSE is dominated by the harder-to-predict high- $T_c$  tail and suggests that the dataset may contain multiple underlying material regimes with distinct superconducting behavior.

A two-dimensional PCA projection of the standardized feature matrix explained approximately 49.4% of total variance (PC1: 38.9%, PC2: 10.5%). Materials with higher critical temperatures tended to occupy specific regions of this projection rather than being uniformly distributed, indicating that the engineered descriptors capture structure related to critical temperature, although the moderate explained variance implies this two-dimensional view captures only part of the dataset’s complexity.

### 3.6 Duplicate and Leakage Robustness

A large fraction of the dataset consists of exact-duplicate feature vectors: 6,093 of 21,263 rows (28.7%) are exact duplicates of another row in feature space. Removing these duplicates leaves 15,170 unique rows. Retraining Random Forest on this deduplicated dataset lowers performance from  $R^2 = 0.930$  (RMSE 8.97 K) to  $R^2 = 0.911$  (RMSE 10.01 K). Because this check removes only *exact* duplicates and not the many additional *near*-duplicate compositions, the true leakage-controlled performance is likely somewhat lower still, and the deduplicated figure should be regarded as the more conservative and honest estimate.

## 4. Discussion and Limitations

The results indicate that engineered elemental descriptors carry substantial predictive signal for critical temperature, and that tree-based ensemble methods — particularly Random Forest and XGBoost — capture this signal considerably better than a linear model and, on this dataset, better than a conventional deep neural network. However, several limitations should be considered when interpreting these results.

- The dataset uses engineered statistical descriptors rather than direct experimental or first-principles physical measurements, which limits the physical interpretability of the learned relationships.
- A large fraction of rows are exact-duplicate feature vectors (28.7%), and many more are near-duplicates. A random split therefore leaks similar compositions across train and test and can optimistically bias reported performance; the deduplicated result in Section 3.6 partially quantifies this effect, but a fully composition-grouped split would be required to remove it entirely.
- Feature importance and correlation analysis identify statistically predictive descriptors but do not establish causal physical mechanisms, particularly given the strong multicollinearity among thermal-conductivity-related features.
- Hyperparameters for the boosting methods and the neural network were only lightly tuned, so the relative ranking among the top models may shift under more systematic tuning.
- PCA is a linear dimensionality-reduction technique used here purely for visualization; it does not provide a direct physical interpretation of the principal components.

For these reasons, the present results should be regarded as a proof of concept rather than a definitive scientific conclusion about the physical determinants of superconducting critical temperature.

## 5. Future Work

- Testing the pipeline on additional, independently sourced superconductivity datasets.
- Using composition-grouped or deduplicated splitting to obtain leakage-free performance estimates.

- Combining engineered descriptors with more direct physical or first-principles measurements (e.g., density-functional-theory-derived features).
- Systematic hyperparameter tuning for all models, including the neural network architecture.
- Developing a ranking-based screening pipeline to prioritize candidate materials for experimental investigation.
- Adding uncertainty estimation (e.g., quantile regression or ensemble variance) to model predictions.

## 6. Conclusion

This study demonstrates that machine learning models can predict superconducting critical temperature from engineered material descriptors with meaningful accuracy. Among six compared models, Random Forest achieved the best performance (cross-validated RMSE  $\approx 9.70$  K,  $R^2 \approx 0.930$ ), narrowly outperforming XGBoost, while a deep multilayer perceptron placed in the middle of the field — confirming that gradient-boosted and bagged tree ensembles remain strong baselines for tabular materials data. The magnitude of improvement over the linear baseline indicates that the relationship between engineered descriptors and critical temperature is strongly nonlinear. Impurity-based and permutation feature importance, correlation analysis, and PCA visualization consistently point to thermal-conductivity-, valence-, mass-, and radius-related descriptors as informative predictors, though these findings should be interpreted as statistical associations rather than confirmed physical mechanisms. A duplicate analysis further shows that some of the headline accuracy reflects near-duplicate compositions shared across the split, so the leakage-controlled estimate ( $R^2 \approx 0.911$ ) is the more conservative figure. Overall, this work serves as a proof-of-concept pipeline for machine-learning-assisted superconductor screening and provides a foundation for future extensions toward more physically grounded materials discovery workflows.

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